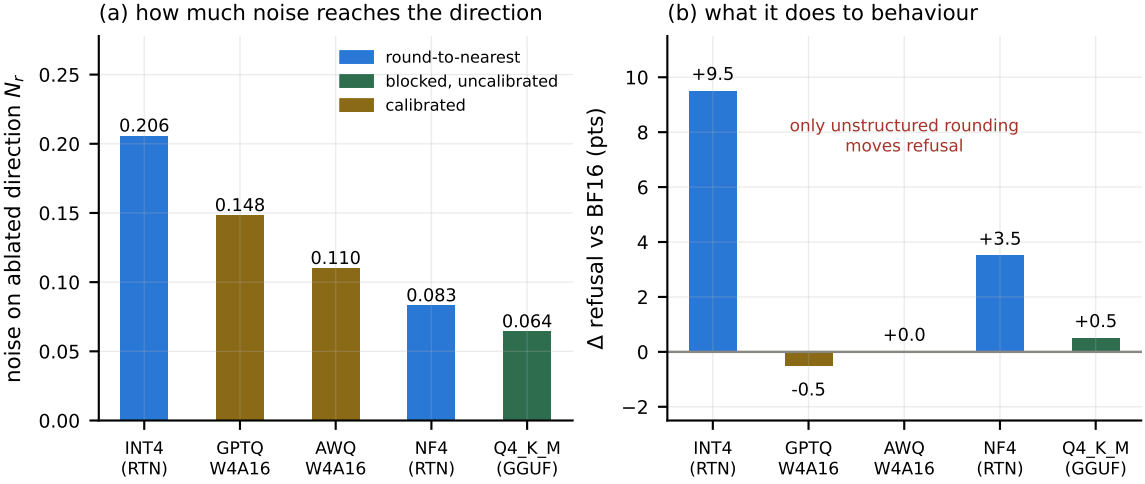
(a) how much noise reaches the direction
round-to-nearest
blocked, uncalibrated
calibrated
noise on ablated direction N_r
0.25
0.20
0.15
0.10
0.05
0.00
0.206
0.148
0.110
0.083
0.064
INT4 (RTN)
GPTQ W4A16
AWQ W4A16
NF4 (RTN)
Q4_K_M (GGUF)
(b) what it does to behaviour
Δ refusal vs BF16 (pts)
10
8
6
4
2
0
−2
+9.5
only unstructured rounding moves refusal
-0.5
+0.0
+3.5
+0.5
INT4 (RTN)
GPTQ W4A16
AWQ W4A16
NF4 (RTN)
Q4_K_M (GGUF)